

Recruitment Training Management System (RTMS)

1. Project Overview

Objective

Build a web-based Recruitment Training Management System where:

- Candidates enroll in training courses.
 - Admin manages courses, batches, trainers, and students.
 - Trainers conduct virtual training sessions through Google Meet.
 - Candidates receive reminders and join sessions through the portal.
 - Session notes, materials, and attendance are maintained.
 - Entire system should work using free/open-source technologies.
 - AI features are not required in V1 but architecture should support future AI integration.
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2. User Roles

Super Admin

Responsibilities:

- Manage Trainers
 - Manage Candidates
 - Create Courses
 - Create Batches
 - Assign Trainers
 - Monitor Sessions
 - Upload Global Materials
 - View Reports
-

Trainer

Responsibilities:

- View Assigned Batches

- Schedule Sessions
 - Add Google Meet Links
 - Upload Session Notes
 - Upload Materials
 - Mark Session Completion
-

Candidate

Responsibilities:

- Login
 - View Courses
 - View Batch Details
 - View Upcoming Sessions
 - Join Google Meet
 - Download Notes
 - Download Materials
-

3. Technology Stack

Frontend

- ReactJS
 - React Router
 - Axios
 - Material UI
 - React Hook Form
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Backend

- Python 3.12+
- FastAPI
- SQLAlchemy
- Pydantic

- JWT Authentication
-

Database

- PostgreSQL
-

Scheduler

- APScheduler

Purpose:

- Meeting reminders
 - Daily notifications
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Email

- Gmail SMTP

Purpose:

- Enrollment emails
 - Session reminders
-

Storage

Local Storage

Example:

```
uploads/
├── notes/
├── materials/
├── profile/
└── assignments/
```

4. System Modules

Module 1: Authentication

Features:

- Login
- Logout
- Forgot Password
- Change Password
- JWT Token Authentication

Tables:

users

Fields:

id

name

email

password_hash

role

is_active

created_at

updated_at

Roles:

ADMIN

TRAINER

CANDIDATE

Module 2: Candidate Management

Admin can:

- Add Candidate
- Edit Candidate
- Deactivate Candidate

Candidate Profile:

id

name

email

phone

education

experience

resume_path

status

Status:

ACTIVE
INACTIVE

Module 3: Trainer Management

Admin can:

- Create Trainer
- Assign Trainer

Trainer Fields:

id
name
email
phone
specialization
status

Module 4: Course Management

Admin can create course.

Course Fields:

id
course_name
description
duration_months
status
created_at

Example:

Python Full Stack
Data Science
GenAI Engineering

Module 5: Batch Management

Purpose:

Group students under trainer.

Batch Fields:

id
batch_name
course_id
trainer_id
start_date
end_date
status

Status:

UPCOMING
ACTIVE
COMPLETED

Example:

Python Batch A
Python Batch B

Module 6: Candidate Enrollment

Purpose:

Assign candidate to course and batch.

Table:

candidate_enrollments

Fields:

id
candidate_id
course_id
batch_id
enrollment_date
status

Status:

ENROLLED
COMPLETED
DROPPED

Module 7: Session Management

Trainer creates training session.

Fields:

id
batch_id
title
description
session_date
start_time
end_time
meet_link
status
created_by

Status:

UPCOMING
COMPLETED
CANCELLED

Workflow:

Trainer

↓

Create Session

↓

Add Meet Link

↓

Save Session

↓

Candidates See Session

Module 8: Google Meet Integration (V1)

No API integration.

Trainer manually creates Google Meet.

Example:

<https://meet.google.com/xyz-abc>

Paste link during session creation.

Store link in database.

Candidate joins using dashboard.

Future Version:

Google Calendar API

Automatic Meet Generation

Module 9: Notification System

Purpose:

Send reminders.

APScheduler Jobs:

1 Day Before Session

1 Hour Before Session

15 Minutes Before Session

Notification Channels:

Email

Workflow:

Scheduler

↓

Check Upcoming Sessions

↓

Fetch Candidates

↓

Send Email Reminder

Module 10: Candidate Dashboard

Sections:

My Courses

Upcoming Sessions

Completed Sessions

Meeting Notes

Study Materials

Attendance

Profile

Module 11: Trainer Dashboard

Sections:

Assigned Batches

Upcoming Sessions

Completed Sessions

Upload Notes

Upload Materials

Candidate List

Module 12: Session Notes Management

After session completion:

Trainer uploads:

- PDF
- DOCX
- PPT

Table:

session_notes

Fields:

id

session_id

title

file_path

uploaded_by

created_at

Workflow:

Training Complete

↓

Trainer Uploads Notes

↓

Candidate Downloads Notes

Module 13: Study Materials

Purpose:

Store training content.

Files:

PDF

PPT

DOCX

ZIP

Fields:

id

course_id

title

description

file_path

uploaded_by

created_at

Module 14: Attendance

Simple Attendance System.

When candidate clicks:

Join Meeting

System creates record.

Table:

attendance

Fields:

id

candidate_id

session_id

join_timestamp

Future:

Google Meet Attendance Integration

Module 15: Reports

Admin Reports

Total Students

Total Trainers

Active Courses

Active Batches

Upcoming Sessions

Completed Sessions

5. Database Schema

Core Tables:

users

candidates

trainers

courses

batches

candidate_enrollments

sessions

session_notes

materials

attendance

notifications

6. Backend Folder Structure

backend/

app/

├── auth/

├── users/

├── candidates/

├── trainers/

├── courses/

├── batches/

├── enrollments/

├── sessions/

├── notes/

├── materials/

└── attendance/

- ├── notifications/
- ├── scheduler/
- ├── database/
- ├── middleware/
- ├── utils/
- └── main.py

7. Frontend Folder Structure

frontend/src

- ├── pages/
 - | ├── Login
 - | ├── Dashboard
 - | ├── Courses
 - | ├── Batches
 - | ├── Sessions
 - | ├── Notes
 - | └── Materials
- ├── components/
- ├── services/
- ├── hooks/
- ├── layouts/
- ├── routes/
- └── utils/

8. API Design

Auth APIs

POST /auth/login

POST /auth/logout

POST /auth/change-password

Candidate APIs

GET /candidates

POST /candidates

PUT /candidates/{id}

DELETE /candidates/{id}

Trainer APIs

GET /trainers

POST /trainers

PUT /trainers/{id}

DELETE /trainers/{id}

Course APIs

GET /courses

POST /courses

PUT /courses/{id}

DELETE /courses/{id}

Batch APIs

GET /batches

POST /batches

PUT /batches/{id}

DELETE /batches/{id}

Session APIs

GET /sessions

POST /sessions

PUT /sessions/{id}

DELETE /sessions/{id}

Notes APIs

POST /notes/upload

GET /notes/{session_id}

Materials APIs

POST /materials/upload

GET /materials

Attendance APIs

POST /attendance/join

GET /attendance/session/{id}

9. Deployment

Development

Frontend:

React

Backend:

FastAPI

Database:

PostgreSQL

Production (Free)

Frontend:

Vercel

Backend:

Render

Database:

Supabase PostgreSQL

File Storage:

Server Local Storage

SMTP:

Gmail SMTP

10. Future Scope (Not in V1)

- Automatic Google Meet Creation

- WhatsApp Notifications
 - AI Meeting Summary
 - AI Attendance Analysis
 - AI Chatbot
 - Certificate Generation
 - LMS Features
 - Assignment Submission
 - Online Assessment
 - Mobile Application
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Final MVP Deliverables

1. Authentication & Authorization
2. Candidate Management
3. Trainer Management
4. Course Management
5. Batch Management
6. Candidate Enrollment
7. Session Scheduling
8. Google Meet Link Management
9. Email Reminder System
10. Candidate Dashboard
11. Trainer Dashboard
12. Notes Upload & Download
13. Study Material Management
14. Attendance Tracking
15. Reports Dashboard

Goal: A complete Training Management System where candidates can enroll, join trainer-led Google Meet sessions, receive reminders, access notes and materials, and track their training progress through a centralized portal without using any paid services.